Md Nadim Mahmud

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Research Interests

My research focuses on machine learning, generative AI, and computer vision. Particularly energy-efficient ML pipelines that integrate multimodal sensing and foundation models for real-world edge deployment, with applications spanning human-centered AI, accessible navigation, and assistive technology.

Education

Miami University — Oxford, OH, USA
M.S. in Computer Science — *Aug. 2024 – May 2026*

- GPA: 3.94/ 4.00
- Graduate Assistantship: Full tuition scholarship and living stipend
- Thesis: MyPath: An AI-Powered Accessibility Assistant for Wheelchair Navigation
- Research Interests: LLM, AI, Machine Learning, mmWave Radar, Wheelchair Accessibility

University of Rajshahi — Rajshahi, Bangladesh
BSc. in Computer Science & Engineering — *Jan 2017 – May 2022*

- GPA: 3.92 / 4.00 (Top of the batch)

Publications

[1] N. Mahmud, V. Raychoudhury, M. O. Gani, and S. Saha, *Surfaceencoder: Semi-supervised representation learning for global wheelchair accessibility*, Accepted at ECML PKDD, 2026.

[2] S. Karki, E. Han, N. Mahmud, S. Bhunia, J. Femiani, and V. Raychoudhury, "Omniacc: Personalized accessibility assistant using generative ai," in *AAAI 2025, AI for Urban Planning*, arXiv preprint arXiv:2509.07220, 2025.

[3] A. S. M. Hossain, N. Mahmud, M. O. Gani, and V. Raychoudhury, "Omnipath: A multi-modal agentic framework for auditing wheelchair accessibility," in *Proceedings of the IEEE 50th Annual Computers, Software, and Applications Conference (COMPSAC)*, To appear, IEEE, 2026.

[4] A. M. Hossain, N. Mahmud, E. Han, *et al.*, "A smart and barrier-aware navigation system for wheelchair routing," in *2025 IEEE 22nd International Conference on Mobile Ad-Hoc and Smart Systems (MASS)*, 2025, pp. 717–722. DOI: 10.1109/MASS66014.2025.00117.

[5] E. Han, N. Mahmud, C. Cordon, *et al.*, *Beyond noisy ground truth: Noisily supervised semantic segmentation of accessible sidewalks and crosswalks from satellite images*, Accepted at IEEE SmartComp, 2026.

Achievements and Programming Contests

- MU Summer Research Fellowship Awardee, Miami University — Summer 2025, 2026
- Faculty of Engineering Dean's List, University of Rajshahi — Academic Years: 2017, 2018, 2019, 2020
- Ranked 6th in the ITEE FE Bangladesh Exam taken by IPA of Japan — Nov 2021
- Passed professional test at Samsung R&D Bangladesh — Sept 2022
- Ranked 25th, Nadim_Mahmud, Samsung R&D Bangladesh Coding Contest — Sept 2019
- Ranked 74th, RU_Squadron, ICPC Dhaka regional contest — 2019

Research Experience

Graduate Assistant — Aug. 2024 – May 2026
Miami University — *Oxford, OH*

- Accessible routing, Navigation path generation, Zero-shot learning, mmWave-based feature detection
- Applied zero-shot learning using the OpenAI API for recognizing sidewalks as accessibility features
- Working with mmWave radar–based methods for road feature detection
- Contributed to the preparation of NIH R15 grant writing
- Mentored a group of four undergraduate students on their capstone project

- Reviewed papers for ECML PKDD

Undergraduate Researcher Aug 2021 – May 2022
University of Rajshahi *Rajshahi, Bangladesh*

- **Thesis:** A Machine Learning Approach for Differentiating Resting State Neural Activity in Autism Spectrum Disorder (ASD) from Control using rs-fMRI
- Research on fMRI images of different Autism Spectrum Disorder patients & find out the correlation among the resting state brain function of different regions of interest
- Implemented data pre-processing script for ROI-extraction, correlation & union of voxels of each ROI

Employment

Software Engineer Oct 2022 – July 2024
Therap Bd Ltd. (Multinational) *Dhaka, Bangladesh*

- **Tech Stack:** Java Spring Framework, Hibernate, Oracle SQL, JSP, JavaScript, jQuery
- Played a key role in developing a scalable push notification framework, with a 300,000+ user base
- Developed Excel upload/download solutions using internal frameworks, improving data entry efficiency by 80%
- Implemented advanced archive and search functionality with internal frameworks
- Developed database migration scripts
- Conducted thorough code reviews, ensuring quality and best practices

Leadership and University Engagements

Graduate Teaching Assistant Oct 2024 – Present
Miami University *Oxford, OH*

- Mentoring students in their Software Engineering projects, helping with design and implementation
- Mentoring projects for the course Intro to AI

Member of Faculty Learning Center - AI & Student Project Development Fall 2025 – Spring 2026
Miami University *Oxford, OH*

- Exploring and evaluating AI tools (e.g., ChatGPT, Grammarly) for academic writing
- Will develop guidelines and AI-integrated lesson plans, collaborating on departmental recommendations and alternative assessment strategies to support responsible AI use in coursework.

Capstone Project Mentor Fall 2025 – Spring 2026
Miami University *Oxford, OH*

- Mentoring 4 undergraduate students on their capstone research projec for blind people navigation

Programming Club Trainer Jan 2018 – Nov 2021
University of Rajshahi *Rajshahi, Bangladesh*

- Conducted 100+ Algorithm and Data Structure sessions
- Organized 20+ programming contests

Projects

MyPath – Wheelchair-Accessible Routing App | *JavaSpring, Hibernate, PostgreSQL* Dec 2024 – Present

- Developed a Java Spring Boot API to generate wheelchair-accessible routes based on latitude-longitude queries
- Returns routes with surface type, incline, turns, and duration for optimized accessibility

Deep Learning-Based Learned Skeletonization of Road Networks | *CNN* April 2025 – Present

- Developed a deep learning model to recover thin road skeletons from thick, noisy OpenStreetMap (OSM) images
- Training a custom U-Net with weighted focal loss and residual blocks, achieving high accuracy on distorted, rasterized road data

Multivendor E-Commerce | *JavaSrping, PostgreSQL, Hibernate* Jan 2023 – April 2023

- WebsiteImplemented user and product management, shopping cart functionality, and order processing.
- Utilized Java, Spring, PostgreSQL, Hibernate, and JSP with role-based authentication (Admin, User, Shopkeeper, Deliveryman).
- Automated project build and deployment using Gradle and Docker.

RU Contacts App (10000+ downloads) | *Kotlin, MVVM* April 2022 – June 2022

- Developed a contacts app for Rajshahi University using Android Studio and Kotlin
- Implemented MVVM architecture for better maintainability and scalability

Technical Skills

Languages Java Spring, Hibernate, JSP , MySQL, Oracle SQL
Frameworks: React, Node.js, Flask, JUnit, WordPress, Material-UI, FastAPI
Developer Tools: Git, Docker, VS Code, IntelliJ
Libraries: pandas, NumPy, Matplotlib